Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void func(int n)
{
    int count = 0;

    count++;                    // if condition

    if (n == 1)
    {
        count++;                // printf("*")
    }
    else
    {
        for (int i = 1; i <= n; i++)
        {
            count++;            // outer loop execution

            for (int j = 1; j <= n; j++)
            {
                count++;        // inner loop execution
                count++;        // printf("*")
                count++;        // printf("*")
                break;
            }

            count++;            // outer loop increment
        }

        count++;                // final condition
    }

    printf("%d", count);
}
```

```c
int main()
{
    int n;
    scanf("%d", &n);

    func(n);

    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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